

Final Review Session Spring 2026

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

**Recap - PPDAC/study
design/variable types**

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recap - PPDAC/study design/variable types

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Types of questions:

Remember we discussed 3 general types of questions:

- ▶ Descriptive
- ▶ Predictive
- ▶ Causative/etiologic

For which type of question are we generally setting up hypothesis testing?

Types of studies

We want to think about important aspects of the study that determine how the data might be used:

- ▶ Observational vs Experimental
- ▶ Temporality:
 - ▶ prospective
 - ▶ retrospective
 - ▶ cross- sectional
- ▶ If experimental - details:
 - ▶ controlled
 - ▶ randomized (at what level)
 - ▶ blinding?
- ▶ Selection/representativeness
 - ▶ census
 - ▶ random selection
 - ▶ proportional/probabilistic selection
 - ▶ conditional selection (conditional on exposure or conditional on outcome)
 - ▶ non-probability samples

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Types of variables

Quantitative: - continuous - discrete

Categorical: - ordinal - nominal

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recap - PPDAC/study
design/variable types

**Reminder - probability
language**

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Reminder - probability language

Recap - PPDAC/study
design/variable types

**Reminder - probability
language**

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

The probability of having arthritis is 7%. We will denote $P(A)$ as the probability of having arthritis in the population and $P(D)$ as the probability of having a diagnosis of reduced mobility. The probability of having a diagnosis of reduced mobility among those with arthritis is 35%. The probability of not having a diagnosis of reduced mobility and also not having arthritis is 74%.

Independence, dependence, disjoint

Recap - PPDAC/study
design/variable types

**Reminder - probability
language**

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recap - PPDAC/study
design/variable types

Reminder - probability
language

**Recap - confidence intervals
and testing**

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recap - confidence intervals and testing

How confidence intervals behave

Recap - PPDAC/study
design/variable types

Reminder - probability
language

**Recap - confidence intervals
and testing**

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recall the form of a CI:

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

Where $z^* \frac{\sigma}{\sqrt{n}}$ is the **margin of error**.

The margin of error gets smaller when:

- ▶ z^* is smaller (i.e., you change to a smaller confidence level). Thus, there is a trade-off between the confidence level and the margin of error.
- ▶ σ is smaller. You might be able to reduce σ if there is measurement error. Often times, the σ can't be reduced, it is just a characteristic of the population
- ▶ n is larger.

How hypothesis tests behave

- ▶ Statistical significance depends on sample size (since sample size determines the standard error of the sampling mean)
- ▶ Recall the form of the z-test:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{\text{magnitude of observed effect}}{\text{size of chance variation}} = \frac{\text{signal}}{\text{noise}}$$

- ▶ The numerator quantifies the distance between what you observe in the sample and the null hypothesized parameter.
- ▶ The denominator represents the size of chance variations from sample to sample

How hypothesis tests behave

Recap - PPDAC/study
design/variable types

Reminder - probability
language

**Recap - confidence intervals
and testing**

Overview of Part III

Some examples - what
tests?

Additional examples for
review

- ▶ Statistical significance depends on:
 - ▶ The size of the observed effect ($\bar{x} - \mu$)
 - ▶ The variability of individuals in the population (σ)
 - ▶ The sample size (n)
 - ▶ Your criteria for rejection the null (α)

If you obtain a small p-value it is not necessarily because the effect size is large.

Power, Type I error, and Type II error

- ▶ The power is the chance of making the correct decision when the alternative hypothesis is true.
- ▶ Thus, it is the complement of β
- ▶ Power = $1 - \beta$

	H_a is true	H_0 is true
Reject H_0	Correct decision	Type I error (α)
Fail to reject H_0	Type II error (β)	Correct decision

However, there are an infinite number of possible values that μ could assume that are not $= \mu_0$

Thus we must choose a value at which to evaluate the β and power for an alternative hypothesis. . .

When we evaluate β we do so at a single such value μ_1

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Overview of Part III

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Our overarching goal in part III is really to take our two “recipes” for statistical inference

- 1) Hypothesis testing
- 2) Confidence intervals

and figure out which ingredients to add in different situations.

choose the ingredients

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

We want to answer the questions:

- 1) What kind of an outcome variable are we working with?
- 2) How many groups/categories do we have data from that we want to compare?
- 3) Are the groups independent from each other or are observations inherently related/paired?
- 4) Do we meet the assumptions for a parametric test?

choose the ingredients

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Based on the answers to the previous questions, we choose our “ingredients”

- 1) the effect/difference we want to examine
- 2) the variability we have in the data
- 3) a distribution we will be using to draw a critical value from based on:
 - ▶ our desired alpha
 - ▶ one or two tailed hypothesis

Decision tree:

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Decision tree:

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Note: the prezzi presentation also linked on the resources page may be helpful

Recipe 1: Hypothesis testing

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

- 1) Check the assumptions for the theoretical distribution
- 2) Create a ratio of the effect/difference to the variability in the data
- 3) Generate the probability of observing that difference or greater if the null is true
- 4) Make a decision to reject or not reject the null hypothesis

Recipe 2: Confidence intervals

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

- 1) Generate your estimate
- 2) Calculate the critical value associated with your theoretical distribution
- 3) multiply the critical value by the variability
- 4) create your upper and lower bounds

For each test know:

- ▶ When to use it
- ▶ Any important assumptions that can be checked using data
- ▶ Appropriate visualization for the data
- ▶ What theoretical distribution are we using for inference
- ▶ How to construct the statistical test
 - ▶ what are the null and alternative hypotheses for the test
- ▶ How to construct the confidence interval
- ▶ relevant syntax in R
- ▶ any special notes/considerations

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

for example:

One sample T - used when we have 1 sample of a continuous outcome that we are comparing to a hypothesized value - assuming SRS, normality of the outcome, independence of outcomes - we might look at a histogram, density plot or qq plot - compared to a t distribution with $n-1$ df

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

for example:

One sample T - null: the mean is = hypothesized mean - alternative: could be one or two sided

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

for example:

One sample T relevant syntax:

```
pt() t.test(variable, alternative = " ", mu=)
```

Notes: think about when the test is robust to violations of the assumptions and why

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

Some examples - what tests?

Example: what test?

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Defne is working for a research group that wants to compare levels of a potential biomarker for chronic traumatic encephalopathy between a group of former NFL players and a group of controls. The biomarker ranges continuously from 0.70 to 1.60. They are able to recruit 40 former NFL players, and 80 controls.

What is this study design? What test should Defne use and what is her null hypothesis?

Example: what test?

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

A 2023 public health study investigated self-reported pain levels among patients in three different U.S. counties affected by long-term environmental disasters: one exposed to lead-contaminated water (County A), one recovering from a hurricane (County B), and one with ongoing industrial air pollution (County C). Participants were asked to rate their pain intensity on a 0–10 scale over the past week. The researchers noted that the pain scores were heavily skewed right in all counties, with a large number of participants reporting zero pain and a long tail of high-pain outliers. If researchers are interested in testing whether the median pain scores differ across the three counties, what statistical test should they employ?

Today's fun fact

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

The length of your hand from your wrist to your elbow is the same as the length of your foot from your heel to your big toe. Let's say I have access to a medical records system with measurements for all pediatric appointments that includes these measures for over 10,000 children.

What kind of a study design is this? What kind of variables do I have?

If I want to show that these two measures are almost perfectly correlated how might I do that?

Today's fun fact

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

What kind of plot would I expect to see for these data?

What numeric summary measure(s) would be appropriate here?

Example 1: Staph infections

Researchers recruited 917 patients who had tested positive for staphylococcus Aureus and randomly assigned them to a staph-killing nasal ointment or placebo. They were interested in testing whether this drug was associated with a reduction in post-surgical infections. In the active treatment group 17 of 504 patients developed infections, in the placebo group 32 of 413 patients developed infections.

- ▶ What is the study design here?
- ▶ What are the exposure and outcome variables?
- ▶ What kind of a test would you use for these data?
- ▶ What is the null hypothesis of this test?

Recap - PPDAC/study design/variable types

Reminder - probability language

Recap - confidence intervals and testing

Overview of Part III

Some examples - what tests?

Additional examples for review

Staph infections: ingredients

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

- 1) the effect/difference we want to examine
- 2) the variability we have in the data
- 3) a distribution we will be using to draw a critical value from based on:
 - ▶ our desired alpha
 - ▶ one or two tailed hypothesis

Staph infections ingredient 1: effect

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

Difference between two proportions:

$$\hat{p}_1 - \hat{p}_2 = \frac{17}{504} - \frac{32}{413} = .0337 - .0775 = -.0438$$

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

- ▶ If the null hypothesis is true, then p_1 is truly equal to p_2 . In this case, our best estimate of the underlying proportion that they are both equal to is

$$\hat{p} = \frac{\text{no. successes in both samples}}{\text{no. individuals in both samples}} = \frac{17 + 32}{504 + 413} = 0.0534$$

Staff infections ingredient 2: variability

- Our best guess at the SE for $\hat{p}$ is:

$$\sqrt{\frac{\hat{p}(1 - \hat{p})}{n_1} + \frac{\hat{p}(1 - \hat{p})}{n_2}}$$
$$\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

This is the formula for the SE for the difference between two proportions but we have substituted $\hat{p}$ for p_1 and p_2 .

$$\sqrt{0.0534 * (0.9466)\left(\frac{1}{504} + \frac{1}{413}\right)} = 0.01492$$

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Staph infections ingredient 3: distribution

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Here for two sample testing, we have more than 10 successes and 10 failures in each group so we feel comfortable using a normal approximation to the binomial.

We will use a Z distribution, and an alpha of 0.05

The test is one sided because we are only interested in the left side of the distribution - decreases

Staph infections: test statistic

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

$$z = \frac{.0337 - .07748}{\sqrt{.0534 * 0.9466 \left(\frac{1}{504} + \frac{1}{413} \right)}} = -2.936$$

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

```
## [1] 0.001662372
```

In this case we are only interested in a reduction in infections - so we will only look at the left tail of the distribution.

Staph infections

In R?

```
prop.test(x = c(17,32), # x is a vector of number of successes  
          n = c(504,413), alternative="less" , correct=F) # n is a vector of
```

```
##  
## 2-sample test for equality of proportions without continuity correction  
##  
## data:  c(17, 32) out of c(504, 413)  
## X-squared = 8.5906, df = 1, p-value = 0.00169  
## alternative hypothesis: less  
## 95 percent confidence interval:  
## -1.00000000 -0.01839005  
## sample estimates:  
##      prop 1      prop 2  
## 0.03373016 0.07748184
```

Staph infections

In R?

```
binom.test(x=c(17,32),n=c(504,413), alternative="less")
```

```
##  
## Exact binomial test  
##  
## data: c(17, 32)  
## number of successes = 17, number of trials = 49, p-value = 0.02219  
## alternative hypothesis: true probability of success is less than 0.5  
## 95 percent confidence interval:  
## 0.0000000 0.4738246  
## sample estimates:  
## probability of success  
## 0.3469388
```

Staph infections: Confidence interval

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

$$-.0438 \pm 1.96 \sqrt{\frac{.0337(1 - .0337)}{504} + \frac{.0775(1 - .0775)}{413}} = -.0438 \pm 0.01542$$

95% CI is -0.0592 to - 0.04226

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

August 8, 2019 Vitamin D Supplementation and Prevention of Type 2 Diabetes

BACKGROUND Observational studies support an association between a low blood 25-hydroxyvitamin D level and the risk of type 2 diabetes. However, whether vitamin D supplementation lowers the risk of diabetes is unknown.

Example vit d and diabetes

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

METHODS We randomly assigned adults who met at least two of three glycemic criteria for prediabetes (fasting plasma glucose level, 100 to 125 mg per deciliter; plasma glucose level 2 hours after a 75-g oral glucose load, 140 to 199 mg per deciliter; and glycated hemoglobin level, 5.7 to 6.4%) and no diagnostic criteria for diabetes to receive 4000 IU per day of vitamin D3 or placebo, regardless of the baseline serum 25-hydroxyvitamin D level.

What type of study is this?

What type of variable is the predictor (how many groups)?

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

RESULTS By month 24, the mean serum 25-hydroxyvitamin D level in the vitamin D group was 54.3 ng per milliliter (from 27.7 ng per milliliter at baseline), as compared with 28.8 ng per milliliter in the placebo group (from 28.2 ng per milliliter at baseline). After a median follow-up of 2.5 years, the primary outcome of diabetes occurred in 293 participants in the vitamin D group and 323 in the placebo group (9.39 and 10.66 events per 100 person-years, respectively).

What kinds of tests would you use here for the vitamin D comparison? for the outcome?

Example vit d and diabetes

Other considerations: why might we not see the result we were expecting?

Per the discussion in the article:

“Because vitamin D supplements are used increasingly in the U.S. adult population,²⁹ approximately 8 of 10 participants had a baseline serum 25-hydroxyvitamin D level that was considered to be sufficient according to current recommendations (≥ 20 ng per milliliter) to reduce the risk of many outcomes,^{23,30} including diabetes.⁶ The high percentage of participants with adequate levels of vitamin D may have limited the ability of the trial to detect a significant effect.””

Example vaping

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

A study on the effects of vaping classifies people as “never vapers”, “occasional vapers”, “frequent vapers”. You interview a sample of 150 people in each group and ask a questionnaire to derive a quantitative score (between 0 and 100) on stress levels.

What kind of an outcome is this? What test is appropriate here?

Example amygdala

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

The amygdala is a brain structure involved in the processing of memory of emotional reactions. Ten subjects were shown emotional video clips and non emotional video clips in random order. They then had their memory of the clips assessed. Recall accuracy was scored from 1 to 100.

What type of data do you have? What kind of a test is appropriate?

Example births

A random sample of 700 births from local records shows this distribution across the days of the week. Do these data give evidence that local births are not equally likely on all days of the week?

Day	Births
Monday	110
Tuesday	124
Wednesday	104
Thursday	94
Friday	112
Saturday	72
Sunday	84

What test would we use here?

What is the null hypothesis?

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

Example births: expectation

Day	Births
Monday	100
Tuesday	100
Wednesday	100
Thursday	100
Friday	100
Saturday	100
Sunday	100

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

Example grading

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

**Some examples - what
tests?**

Additional examples for
review

You have heard that the grading scale is harsher at UC Berkeley than at other California universities. You want to test this rumor with data. You have data from a random sample of 100 transcripts from students at Berkeley who took PH142 and data on the letter grade distribution for undergraduate statistic courses in general from a California wide survey.

What kind of study is this?

What kind of outcome? How many groups/samples?

What test would you consider here?

Example grading

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

You have heard that the grading scale is harsher at UC Berkeley than at other California universities. You want to test this rumor with data. You have data from a random sample of 100 transcripts from students at Berkeley who took PH142 and data on the letter grade distribution for undergraduate statistic courses in general from a California wide survey.

This is a categorical outcome, with one sample, so we would use a chi-squared goodness of fit test

Example grading

You find a source that gives the distribution for UC intro biostat courses as:
A=50%, B=30%, C=15%, Fail=5%

Grade	N	Expected?
A	224	?
B	99	?
C	17	?
F	10	?
Total	350	350

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Example grading

Grade	N	Expected?
A	224	175
B	99	105
C	17	52.5
F	10	17.5
Total	350	350

statistic= $13.72 + 0.34 + 24.00 + 3.214 = 41.28$

```
pchisq(41.28, df=3, lower.tail=FALSE)
```

```
## [1] 5.703407e-09
```

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

**Additional examples for
review**

Additional examples for review

The melanoma data set contains data on 205 patients from Denmark with malignant melanoma. You have joined a lab in which the principal investigator is interested in determining whether mean tumor thickness (mm) differs by patient status (1 = died from melanoma, 2 = alive, 3 = died from other causes).

```
head(melanoma)
```

##	time	status	sex	age	year	thickness	ulcer	status2
## 1	10	3	1	76	1972	6.76	1	3
## 2	30	3	1	56	1968	0.65	0	3
## 3	35	2	1	41	1977	1.34	0	2
## 4	99	3	0	71	1968	2.90	0	3
## 5	185	1	1	52	1965	12.08	1	1
## 6	204	1	1	28	1971	4.84	1	1

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

```
anova <- aov(thickness ~ status2, data = melanoma)
tidy(anova)
```

```
## # A tibble: 2 x 6
##   term          df sumsq meansq statistic    p.value
##   <chr>      <dbl> <dbl>  <dbl>      <dbl>      <dbl>
## 1 status2         2  180.   90.2        11.3  0.0000216
## 2 Residuals    202 1606.    7.95         NA      NA
```

Your task is to help your PI analyze the results of this ANOVA run in R. For the ANOVA above, what are the null and alternative hypotheses?

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recap - PPDAC/study
design/variable typesReminder - probability
languageRecap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?Additional examples for
review

```
tidy(anova)
```

```
## # A tibble: 2 x 6
##   term          df sumsq meansq statistic    p.value
##   <chr>      <dbl> <dbl>   <dbl>    <dbl>    <dbl>
## 1 status2         2  180.   90.2     11.3 0.0000216
## 2 Residuals    202 1606.    7.95      NA      NA
```

Based on the results shown, what would you conclude?

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Based on these results what would be your next step in the analysis process?
Justify your answer in 1-2 sentences. (would you continue your analysis with an
additional test and if so, what test would you use)

What would you conclude from these results?

```
TukeyHSD(anova)
```

```
##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = thickness ~ status2, data = melanoma)
##
## $status2
##           diff           lwr           upr           p adj
## 2-1 -2.0663511 -3.1192614 -1.013441 0.0000191
## 3-1 -0.5931955 -2.5792549  1.392864 0.7606857
## 3-2  1.4731557 -0.3970052  3.343316 0.1531399
```

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

A group in the athletic department is working with the swim team. They implement a new training program and want to know if there has been an improvement in the 50m swim time (in seconds) at 12 weeks following the start of the program.

Swimming

They have collected the following data:

Pre program	Post Program
24.23	24.26
24.12	24.09
24.15	24.11
24.12	24.13
24.16	24.15
24.18	24.19
24.51	24.42
24.69	24.69
24.88	24.82
25.01	24.94
25.58	25.55
25.47	25.45
25.66	25.67

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Swimming

pre	post	diff	abs	sign	rank
24.69	24.69	0.00	0.00	na	na
24.12	24.13	0.01	0.01	+	2.5
24.16	24.15	-0.01	0.01	-	2.5
24.18	24.19	0.01	0.01	+	2.5
25.66	25.67	0.01	0.01	+	2.5
25.47	25.45	-0.02	0.02	-	5
25.58	25.55	-0.03	0.03	-	7
24.23	24.26	0.03	0.03	+	7
24.12	24.09	-0.03	0.03	-	7
24.15	24.11	-0.04	0.04	-	10
24.88	24.82	-0.06	0.06	-	11
25.01	24.94	-0.07	0.07	-	12
24.51	24.42	-0.09	0.09	-	13

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Calculate the appropriate test statistic. Show your work by writing the formula needed to calculate with values plugged in.

$$Z_T = \frac{T - \mu_T}{\sigma_T}$$

Where

$$\mu_T = \frac{n(n+1)}{4}$$

and

$$\sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{24}}$$

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Write the line of code that would give you the appropriate p-value for this test statistic.:

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

```
pnorm(-1.922, mean = 0, sd = 1)
```

```
## [1] 0.02730288
```

Recap - PPDAC/study
design/variable types

Reminder - probability
language

Recap - confidence intervals
and testing

Overview of Part III

Some examples - what
tests?

Additional examples for
review

Based on your findings would you recommend that the athletic department continue this training program? Why or why not?